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Difference between Twin Screw Extruder and Single screw extruder

Twin-screw extruders and single-screw extruders are both types of extrusion machines used in various industries for processing and shaping materials, particularly in the production of plastic products. Here are the key differences between the two:

1. **Number of Screws:**
   - **Single Screw Extruder:** As the name suggests, it has only one screw that rotates to push the material through the extruder barrel.
   - **Twin-Screw Extruder:** It has two screws that rotate in the barrel simultaneously. The screws can rotate in the same direction (co-rotating) or in opposite directions (counter-rotating).
2. **Mixing Capability:**
   - **Single Screw Extruder:** Generally, it provides less efficient mixing of materials compared to twin-screw extruders. Mixing is primarily achieved through the rotation of the single screw.
   - **Twin-Screw Extruder:** The presence of two screws allows for better mixing of materials. This makes twin-screw extruders more suitable for processes that require thorough blending of multiple components.
3. **Shear and Heat Generation:**
   - **Single Screw Extruder:** It tends to generate more shear heat due to the single screw's rotational movement. This can be a consideration in processes where excessive heat might be detrimental.
   - **Twin-Screw Extruder:** The dual-screw configuration can distribute shear forces more evenly, potentially reducing the overall heat generated during the extrusion process.
4. **Flexibility:**
   - **Single Screw Extruder:** Generally simpler in design and operation, making it more straightforward for certain applications.
   - **Twin-Screw Extruder:** Offers greater flexibility and versatility. It is often used for a wider range of applications, including the processing of materials with varying properties and the incorporation of additives.
5. **Output and Throughput:**
   - **Single Screw Extruder:** In certain cases, single-screw extruders may have limitations on throughput compared to twin-screw extruders.
   - **Twin-Screw Extruder:** Can often achieve higher throughput rates, making them suitable for larger-scale production.
6. **Cost and Complexity:**
   - **Single Screw Extruder:** Typically simpler in design and may be less expensive.
   - **Twin-Screw Extruder:** The dual-screw configuration and increased capabilities may result in higher costs and greater complexity.

Rheological Properties of Materials During the Extrusion Process

The rheological properties of materials play a crucial role in the success of the extrusion process. Understanding how these properties influence the flow, mixing, and shaping of the material is essential for optimizing the process and achieving the desired final product.

Here's a breakdown of key rheological properties and their impact during extrusion:

1. Viscosity:

- **Shear viscosity:** This measures the material's resistance to flow under shear stress. A higher viscosity indicates a thicker, more sluggish material that requires more pressure to extrude. In extrusion, shear rates are typically high, so shear viscosity becomes crucial.
- **Elongational viscosity:** This measures the material's resistance to stretching. It's important for controlling the shape and preventing unwanted necking or thinning of the extrudate.

2. Yield stress:

- This is the minimum stress required to initiate flow. A high yield stress can cause difficulties in starting the extrusion process and lead to uneven flow, especially for low-pressure extrusion.

3. Die swell:

- This refers to the expansion of the extrudate after exiting the die. It's influenced by the material's memory, elasticity, and processing conditions. Excessive die swell can cause dimensional inconsistencies and downstream processing challenges.

4. Slip:

- This refers to the relative movement between the material and the die wall. It can be beneficial or detrimental depending on the desired outcome. Controlled slip can aid in uniform flow and surface quality, while excessive slip can lead to instability and shape irregularities.

5. Time dependence:

- Some materials exhibit shear thinning, where viscosity decreases with increasing shear rate. Others show thixotropy, where viscosity recovers after shearing stops. Understanding these time-dependent behaviors helps predict the material's response to the varying shear rates experienced during extrusion.

Factors affecting rheology:

- **Material type:** Different materials naturally have different inherent rheological properties. Polymers, for example, exhibit complex viscoelastic behavior, while ceramics are typically more rigid and brittle.
- **Temperature and pressure:** Both significantly impact rheology. Extrusion processes often involve elevated temperatures and pressures, which can alter viscosity, yield stress, and other properties.
- **Additives and fillers:** The inclusion of fillers, plasticizers, or other additives can significantly modify the rheology of the base material.

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ADVANTAGE AND DISADVANTAGE OF TWIN SCREW

Advantages:

- High mixing capability - The two intermeshing screws provide high shear and dispersive mixing, allowing for incorporation of fillers/additives and homogenization of blends.
- Continuous processing - Materials are fed into one end and extruded continuously out the other end, allowing for high throughput.
- Versatility - Twin screw designs can be customized for different materials and processes, including compounding, reacting, and shaping operations.
- Tight process control - Extruder barrel sections can be temperature and speed controlled independently for zone-by-zone optimization.
- Low temperature operation - Frictional heat from twin screws allows lower temp extrusion of heat-sensitive materials compared to single screw.
- No melt seals/valves needed - The intermeshing screws form self-wiping seals, preventing backflow.

Disadvantages:

- Higher complexity - More complex equipment requiring tighter tolerances drives higher machine and maintenance costs.
- Material overheating - Excessive heat generation can occur which degrades thermally sensitive materials.
- Screw wear - Abrasion of the screws, barrels, and metal filter elements reduces part life.
- Higher energy consumption - The greater slipping between polymer and screws increases energy usage.
- Shear/extensional heating - Can cause too rapid viscosity drop or degradation of shear-sensitive polymers.

Chemical and nutritional changes in food during extrusion

CHEMICAL CHANGES

1. Gelatinization: The high heat and moisture conditions during extrusion cause starches to gelatinize, making them more easily digestible. This can result in a higher glycemic index for the final product.
2. Maillard Reaction: The extrusion process can lead to the formation of new compounds through the Maillard reaction, a non-enzymatic browning reaction between amino acids and reducing sugars. These compounds can contribute to the development of flavors, aromas, and colors in the final product.
3. Protein denaturation: The high temperatures and pressures during extrusion can cause proteins to denature, or unwind, which can affect their functional properties and bioavailability. Denatured proteins may be less effective at providing structure and texture to the final product.
4. Vitamin loss: The heat and water present during extrusion can lead to the degradation of vitamins, particularly water-soluble vitamins like B and C. This can result in lower levels of these vitamins in the final product.
5. Mineral retention: The extrusion process can help retain minerals in the food by binding them to other components, such as protein or fiber. However, excessive heat can also cause minerals to leach out of the food.
6. Fiber modification: Extrusion can break down or modify dietary fibers, potentially making them easier to digest. However, this can also reduce the overall fiber content of the final product.

7. Lipid oxidation: High temperatures during extrusion can lead to lipid oxidation, resulting in off-flavors and rancidity. Antioxidants can help prevent this from occurring.
8. Moisture content: The extrusion process can influence the moisture content of the final product, with higher moisture levels potentially leading to spoilage and microbial growth.
9. Texture changes: Extrusion can create textural changes in foods, such as puffing, expanding, or crunchifying. This can enhance consumer acceptability but may also affect the nutritional value of the product.
10. Bioactive compound retention: Some bioactive compounds, such as polyphenols, can be preserved or even concentrated during extrusion, depending on the specific processing conditions.

NUTRITIONAL CHANGES\

1. Improved Nutrient Bioavailability: Extrusion can increase the bioavailability of some nutrients by breaking down cell walls and making them more accessible to the body. This can positively affect the nutritional value of the food product.

Impact: The increased bioavailability of nutrients can improve the overall nutritional quality of the food, potentially leading to better health outcomes for consumers.

2. Reduced Anti-Nutritional Factors: Extrusion can reduce the levels of anti-nutritional factors such as phytic acid, which can interfere with nutrient absorption. Lower levels of anti-nutritional factors can improve digestibility and nutrient absorption.

Impact: By reducing anti-nutritional factors, extrusion can make the food more easily digestible and increase the availability of essential nutrients for the body.

3. Loss of Heat-Sensitive Nutrients: Some vitamins and antioxidants are sensitive to heat and may be lost during the extrusion process. Careful processing is necessary to minimize the loss of these nutrients.

Impact: While extrusion can result in some loss of heat-sensitive nutrients, careful processing techniques can help mitigate this loss and preserve the nutritional content of the food.

4. Increased Protein Digestibility: Protein denaturation during extrusion can result in improved protein digestibility, enhancing the utilization of protein in the body.

Impact: The increased digestibility of protein can improve the overall nutritional quality of the food and contribute to better muscle growth and repair.

5. Textural Changes Affecting Perception: Changes in food structure during extrusion can influence its texture, which can impact consumer preference and acceptance.

FACTORS AFFECTING EXTRUSION

1. Temperature: Temperature plays a critical role in the extrusion process. It affects the viscosity of the polymer melt, the rate of melting, and the formation of the extrudate. High temperatures can cause the polymer to degrade or burn, while low temperatures can result in a poorly formed extrudate.
2. Pressure: The pressure applied to the polymer melt also affects the extrusion process. High pressures can cause the polymer to deform and create voids in the extrudate, while low pressures can result in a weak or brittle extrudate.
3. Shear Rate: The shear rate, or the rate at which the polymer melt is forced through the die, affects the viscosity of the melt and the formation of the extrudate. High shear rates can cause the polymer

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to degrade or become discolored, while low shear rates can result in a rough or irregular surface finish.

4. Die Design: The design of the die, including its shape, size, and geometry, can affect the properties of the extruded material. For example, a larger die can produce a wider extrudate, while a smaller die can produce a narrower one. The shape of the die can also affect the cross-sectional profile of the extrudate.
5. Polymer Properties: The properties of the polymer being extruded, such as its molecular weight, crystallinity, and thermal stability, can all affect the extrusion process. For example, a high-molecular-weight polymer may require higher temperatures and pressures to melt and extrude, while a low-crystallinity polymer may have a lower melting temperature but may also be less rigid.
6. Additives: Additives such as pigments, fillers, and plasticizers can affect the properties of the extruded material. Pigments can change the color of the extrudate, while fillers can increase its strength and stiffness. Plasticizers can improve the flexibility of the extrudate, but can also affect its durability and resistance to heat.
7. Screw Design: The design of the screw used in the extruder can also affect the extrusion process. The length, diameter, and pitch of the screw can all influence the mixing and melting of the polymer, as well as the flow of the melt through the die.
8. Barrel Temperature: The temperature of the barrel, which houses the screw and polymer melt, can affect the melting and mixing of the polymer. Uneven heating or cooling along the length of the barrel can result in variations in the properties of the extruded material.
9. Cooling System: The cooling system used after the extrusion process can affect the final properties of the extruded material. Air cooling, water cooling, or a combination of both can be used to control the temperature of the extrudate and prevent warpage or distortion.
10. Process Control: Finally, process control parameters such as speed, pressure, and temperature can all affect the properties of the extruded material. Consistent process conditions and monitoring of the extrusion process can help ensure consistent quality and properties of the extruded material.

GLASS CONTAINERS

- **Ancient Origin:** Glass, used for packaging, has a history of more than 3000 years, starting in Egypt around 1500 B.C.
- **Roman Influence:** During the Roman golden age, glass containers gained popularity for their craftsmanship and aesthetic appeal.
- **Modern Challenges:** Despite competition from newer materials like metal and plastic due to global technological advancements, glass containers remain relevant.
- **Distinct Characteristics:** Glass containers have inherent features such as fragility and tare

Glass Container Properties Simplified:

1. **Chemically Inert:**
   - Glass doesn't undergo chemical reactions with products, ensuring safe packaging without spoilage.
   - Example: No risk of chemical interactions affecting the contents.
2. **Non-Permeable:**
   - Gases, odors, vapors, and liquids cannot penetrate or pass through glass.
   - Example: Prevents leakage or absorption of external elements.
3. **Transparent:**
   - Glass allows clear visibility of the product inside the container.
   - Example: Enables easy inspection of packaged items.
4. **Moldable:**
   - Glass containers can be easily shaped and sized, accommodating various product needs.
   - Example: Available in diverse shapes and sizes for different products.
5. **Strength:**

- Glass exhibits strength comparable to the most robust sheet materials.
- Example: Glass containers provide robust protection for the contents.

6. **Lightweight:**
   - Glass is light, approximately one-third the weight of steel and lighter than aluminum.
   - Example: Facilitates easy handling and transportation due to reduced weight.
7. **Uniform Distribution:**
   - Design modifications ensure even distribution of glass, enhancing container strength.
   - Example: Improved strength without adding extra weight.
8. **Unlimited Supply:**
   - Glass production relies on abundant raw materials like sand, limestone, and soda ash.
   - Example: Accessibility to materials ensures a continuous supply of glass containers.
9. **Raw Material Additions:**
   - Various substances like arsenic, selenium, cobalt oxides, carbon, and sulfides are added for specific glass properties.
   - Example: Boron strengthens glass, while certain elements contribute to color variations.

<u>ADVANTAGES OF GLASS CONTAINERS - SIMPLIFIED:</u>

a) ***Versatile Usage:***

- *Glass containers are suitable for various products like liquids, powders, pastes, and granules.*

b) ***Temperature Resilience:***

- *Glass exhibits both high and low-temperature resistance, making it ideal for products like hot-filled tomato ketchup that can be refrigerated.*

c) ***Non-Contaminating:***

- *Unlike metal or alloy packages, glass containers prevent contamination with crystals or fibers, ensuring purity.*

d) ***Climate Resilience:***

- *Glass containers and their contents remain unaffected by climatic variations.*

e) ***Transparency Benefits:***

- *The see-through property of glass allows easy detection of product color, appearance, and consistency.*

f) ***Customizable Sizes:***

- *Glass containers can be crafted in any size, shape, or capacity to meet specific requirements.*

g) ***Non-Impacting Quality:***

- *Glass containers don't taint, pollute, or affect the quality of the packed product.*

h) ***Diverse Closures:***

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- *Various closures made of metal, plastic, cork, or rubber can be chosen based on specific needs.*

i) ***Impermeable Barrier:***

- *Glass containers provide an impermeable barrier to moisture and gases.*

j) ***Color Options:***

- *Glass containers can be colored (brown, green, blue) to shield products from sunlight.*

APPLICATIONS SIMPLIFIED:

- *Glass containers find wide applications in pharmaceuticals, dairy, liquor, breweries, food, soft drinks, cosmetics, chemicals, inks, and more.*
- *Specifically, glass containers are used for jams, jellies, squashes, pickles, preserves, processed meat and fish products, tea, coffee, spices, baby food, cheese, ghee, and other items.*

METAL CANS - SIMPLIFIED:

Metal cans can be made from aluminum, tin plate, or tin-free steel, but tin plate and aluminum containers are widely used in packaging. Tin plate containers, especially, have been a staple in food packaging for about five decades, making them one of the oldest packaging materials for the food industry after glass containers.

Types of Tin Plate Containers:

a) Open Top Container:

- Standard round containers, also known as Open Top Sanitary (OTS) cans, where customers can choose the size based on their needs.
- The can has a seamed bottom, and the top ends are supplied separately. The food manufacturer fills the can, uses a seamer machine to attach the top lid, and then subjects the filled cans to heat processing.
- Also referred to as a 3-piece can.
- Widely used for processed food and beverages.
- Recent development: Introduction of easy open ends with a "ring pull tab" for effortless opening without a traditional can opener.

b) General Line Container:

- These containers vary in size from small pharmaceutical ointment cans to 15-liter containers for vanaspati or biscuits, and 20-liter drums for paints and varnishes.
- Tailor-made sizes, both round and irregular, are available with various fitments made from tinplate, aluminum, and plastics.
- Primarily used for bakery products, hydrogenated oils, and confectionery items, rather than processed foods.

Recent Development:

- Introduction of DWI cans or "Drawn and Wall Iron" cans, categorized as two-piece cans. The cylindrical body and bottom form a single piece, while the lids are separate. These cans are advantageous for pressurized beer and beverages, offering material savings without compromising machine speed.

Applications of Metal Cans - Varied Usage:

Metal cans, including those made from tin plate, aluminum, or tin-free steel, are versatile and widely used for packaging diverse products. The applications of metal cans include:

1. **Food Packaging:**
    - **Canned Vegetables and Fruits:** Metal cans are commonly used to package fruits and vegetables. The canning process helps preserve the freshness and nutritional value of the produce.
    - **Soups and Stews:** Many ready-to-eat soups and stews are packaged in metal cans, providing a convenient and long-lasting option for consumers.
2. **Beverage Packaging:**
    - **Soft Drinks and Carbonated Beverages:** Aluminum cans are extensively used for packaging carbonated beverages due to their ability to maintain carbonation and protect the contents from light and air.
    - **Canned Beer:** Aluminum cans are a popular choice for packaging beer due to their light weight, recyclability, and ability to prevent light exposure, which can affect the beer's flavor.
3. **Canned Seafood and Meat:**
    - **Tuna, Salmon, and Sardines:** Metal cans are commonly used to preserve and package various types of seafood. The canning process ensures the long shelf life of these products.
    - **Canned Meat Products:** Processed meats such as chicken, beef, and pork are often packaged in metal cans, providing a shelf-stable and convenient option for consumers.
4. **Pet Food Packaging:**
    - **Wet Pet Food:** Metal cans are used to package wet pet food, ensuring the preservation of flavors and nutrients while extending shelf life.
5. **Aerosol Cans:**
    - **Personal Care Products:** Aerosol cans made of metal are commonly used to package products such as deodorants, hairsprays, and shaving creams.
    - **Household Products:** Aerosol cans are also used for packaging various household products like insect repellents, air fresheners, and cleaning sprays.
6. **Chemical and Industrial Products:**
    - **Paints and Coatings:** Metal cans are used for packaging paints and coatings due to their ability to protect the contents from air and light.
    - **Solvents and Chemicals:** Certain chemicals and solvents are packaged in metal cans to ensure safe storage and transportation.
7. **Coffee and Tea Packaging:**
    - **Coffee:** Some brands use metal cans for packaging ground coffee or coffee beans, providing a sealed and airtight container to preserve flavor and freshness.
    - **Tea:** Metal cans are also used to package specialty teas, providing protection against moisture and light.
8. **Confectionery and Snacks:**
    - **Nuts, Candies, and Snack Items:** Metal cans are used for packaging various snacks and confectionery items, providing a durable and protective container.

Types of Metal Cans:

1. **Aluminum Cans:**
    - Lightweight and easily recyclable.
    - Commonly used for beverages such as soft drinks, beer, and energy drinks.
    - Offers excellent protection against light and air.

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2. **Steel Cans:**
    - Durable and strong.
    - Used for a variety of products, including canned vegetables, fruits, soups, and meats.
    - Popular in the food industry due to its ability to withstand high temperatures during processing.
3. **Tinplate Cans:**
    - Steel cans coated with a thin layer of tin to prevent corrosion.
    - Commonly used for packaging food items like canned fruits, vegetables, and dairy products.
4. **Aerosol Cans:**
    - Designed for dispensing products in spray or mist form.
    - Commonly used for personal care items, household products, and industrial sprays.
5. **Bi-Metal Cans:**
    - Combine materials like aluminum and steel for enhanced performance.
    - Often used in the production of beverage cans to leverage the strengths of both materials.
6. **Composite Cans:**
    - Typically made of paperboard with metal ends.
    - Used for products like snacks, powders, and non-carbonated beverages.

Differentiating OTS and DWI Cans:

1. **OTS (Open-Top Sanitary) Cans:**
    - Have a removable metal lid.
    - Suited for products that need to be accessed frequently.
    - Examples include paint cans and some food containers.
2. **DWI (Drawn and Wall-Ironed) Cans:**
    - Made from a single piece of metal shaped into a can.
    - Typically used for beverages and other products where a sealed, airtight container is essential.
    - The seamless design enhances structural integrity.

Main Applications of OTS Cans:

1. **Paint Cans:**
    - Commonly used for packaging paints, coatings, and varnishes.
    - The open-top design allows easy access to the product.
2. **Food Containers:**
    - Some food products, especially those that may require frequent access, are packaged in OTS cans.
    - Examples include certain types of canned fruits, vegetables, and sauces.
3. **Industrial Products:**
    - OTS cans are used for packaging various industrial products and chemicals.
    - The open-top design facilitates easy pouring or scooping.
4. **General Storage:**
    - OTS cans are also used for general storage purposes where the contents need to be accessed regularly.
    - Examples include storage containers for hardware items or small parts.
5. **Customized Packaging:**
    - OTS cans are versatile and can be customized for various products based on specific packaging requirements.

ALUMINIUM FOIL - SUPERIOR FLEXIBLE PACKAGING MATERIAL

Over time, aluminium foil has solidified its position as the premier flexible packaging material, boasting unparalleled barrier properties against moisture, gases, light, and aroma when compared to other materials. The unique characteristics of aluminium foil stem from its ability to retain all the metallic properties of aluminium.

Aluminium foil is a continuous sheet or web of aluminium metal, rolled to thicknesses ranging from 0.005 to 0.2 mm. It is derived from commercially pure aluminium, ensuring an aluminium content of not less than 98 percent. Available in hard or soft temper, with a bright or one side dull/matt surface, it comes in plain, colored, coated, lubricated, and embossed forms.

Key Properties:

1. *Impermeability:* Foil with a thickness of 0.025 mm is impermeable to moisture vapor and gas transmission.
2. *Non-toxic:* Due to its inert nature, aluminium foil is completely non-toxic.
3. *Stable:* It remains stable and does not become brittle at low temperatures.
4. *Light and Heat Barriers:* Acts as a barrier against light and heat.
5. *Tasteless and Odorless:* Does not absorb odors from food nor imparts any off odors to the food.
6. *Tamper-proof:* Functions as a pilfer-proof diaphragm for tin containers.
7. *Retort Pouch:* Used as the middle layer in three-layered flexible laminates to withstand temperature and pressure in the retort, extending the shelf life of ready-to-eat products.

Advantages:

- *Tearing Properties:* Facilitates the use of the web as a sealing surface in blister packs.
- *Impermeable Nature: Extensively used for lamination.*
- ***Blocks Out Stuff:***
- ***Safe and Sound:***
- *Because it doesn't react, it won't harm what it wraps - completely safe.*
- ***Stays Strong in the Cold:***
- *Even in chilly temps, it stays strong and won't break easily.*
- ***Defends Against Light and Heat:***.
- ***No Weird Tastes or Smells:***
- *It won't pick up smells or pass any strange tastes to the food - everything stays just as it should.*

Applications:

1. *Decorative Label:* 0.009 mm foil laminated to paper for flexible labels.
2. *Confectionery:* 0.009 mm foil, wax-backed by paper, for packing chocolates and toffees.
3. *Biscuit Wrappers:* 0.009 mm foil and 30 gsm poster paper for vibrant, wax-coated biscuit wrappers.
4. *Milk Products:* Cheese wrappers composed of 0.012 mm foil coated with heat-sealable, anti-corrosive coatings.
5. *Multilayer Laminates:* Used in packaging for instant coffee, pan masala, snack foods, and aseptic pouches.
6. *Stand-Up Pouches:* Employed for packing ready-to-serve (RTS) beverages.
7. *Tea Chest Lining:* Bulk packaging of tea.
8. *Milk Strip:* Used for capping.

PLASTIC MATERIALS *are solids that can change shape when heated without falling apart. They're made from organic or inorganic compounds with a lot of molecules, and we often call them synthetic resins. These materials are fantastic for packaging because they're light, strong, and don't rust or react with chemicals.*

Plastics can be molded or shaped during manufacturing by using heat and pressure. There are two types: thermoset, which keeps its shape even when heated, and thermoplastic, which can change shape when heated.

In India, plastic use is growing fast because it's easy to find the materials needed. Plastics like polyethylene and polypropylene are commonly used for packaging because they are lightweight, strong, and don't react with food. About one-third of all plastics made are used for packaging, and two-thirds of that is for food and drinks.

Plastics have benefits like being lightweight, strong, flexible, and recyclable. They're also good at keeping moisture and gases away from food, making them ideal for packaging.

In simpler terms, plastics are awesome for packaging because they are versatile, easy to shape, and keep our food fresh and safe.

Polymer Type	Example Materials
Polyolefins	- Polyethylene - Polypropylene (PP)
Polyvinyl group	- Polyvinyl Chloride (PVC)
Condensation polymer	- Polyester (PET)- Nylon-6 or Polyamide (PA)
Styrene polymers	- Polystyrene (PS) - Expanded Polystyrene (EPS)
Carbonate group	- Polycarbonate (PC)

Polyolefins Group: a) **Low-Density Polyethylene (LDPE) Film:** • Density: 0.910 – 0.925 gms/cc • Molecular Weight: 3 × 10^5 • Heat Resistance: 82 – 100°C • Clarity: Translucent • Water Absorption: 0.015% • Gas Permeability: 1.0 cc/m2/24hrs at 27°C and 1 atm pressure • Strength: Good tensile strength, high elongation, good dart impact resistance • Resistance: Resistant to weak acids and alkalies, excellent barrier to moisture vapor	**Linear Low-Density Polyethylene (LLDPE):** • High film tensile strength • High elongation, tear strength, and puncture resistance • Improved stiffness • Excellent heat seal properties

) **Medium-Density Polyethylene (MDPE) Film:** • Density: 0.926 – 0.940 gm/cc • Molecular Weight: 2 × 10^5 • Heat Resistance: 105 – 121°C • Clarity: Translucent • Water Absorption: 0.01 • Gas Permeability: 1.33 cc/m2/24 hrs at 27°C and 1 atm pressure • Resistance: Very resistant to weak acids and alkalies	**Polypropylene (PP) Film:** • High tensile strength • Chemical resistance, high-temperature performance • Low permeability to moisture and gas • High transparency, chemical inertness
High Molecular High-Density Polyethylene Film (HMHDPE): • High mechanical strength • White translucence, high tear resistance • No taste or odor, suitable for food contact • Excellent moisture barrier	**High-Density Polyethylene (HDPE) Film:** • Density: 0.941 – 0.965 gm/cc • Molecular Weight: 1.25 × 10^5 • Heat Resistance: 121°C • Nature: Opaque • Barrier: High barrier to moisture vapor • Gas Permeability: Less compared to other polyethylene films
⍰ **Polyvinyl Chloride (PVC):** • Hard, brittle, transparent material • Low Gas Transmission Rate (GTR), moderate Water Vapor Transmission Rate (WVTR) • Good resistance to fat and oil, glass-like clarity • Excellent printability, chemical resistance	⍰ **Condensation Polymer:** a) **Polyester Film (PET):** • Excellent gloss • Low moisture and gas permeability • High mechanical strength, tear resistance • Dimensionally stable, easy printability b) **Polyamide or Nylon-6 Film:** • High mechanical strength, elongation capability • Resistance to cutting, abrasion, bursting, and chemicals • Outstanding impermeability to gases, easy printability

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⍰ **Styrene Polymers:** a) **Polystyrene (PS):** • Rigid containers, crystal clarity • Attractive colors, lustrous finish • Dimensional stability, resistance to chemicals • Easy processing, good moisture barrier b) **Expanded Polystyrene (EPS):** • Cushioning material, used in packaging • Various forms: rigid sheets, flexible sheets, granules, cushion molded	

PLASTIC CONTAINERS

Plastic containers are manufactured with the help of different processes and

are available in different forms like bottle, drums, carboys, jars, etc. These

have wide application in packaging of processed food products.

The plastic containers are manufactured by means of different processes like:

a) Blow moulding process

b) Extrusion blow moulding process

c) Stretch blow moulding process

d) Injection blow moulding process

e) Coextrusion blow moulding process

Collapsible Containers

Collapsible containers, often found in the form of tubes for packaging, are made from either multilayer plastics or aluminum. While aluminum collapsible tubes have been traditionally used for pharmaceutical products, plastic tubes are gaining popularity for items like toothpaste, ayurvedic medicines, cosmetics, stationery, and more.

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I. Aluminum Collapsible Tube

Manufacturing Process: The extrusion process is used to create collapsible aluminum tubes. After extrusion, the tubes are initially hard and do not collapse. An annealing process follows, imparting softness and removing impurities. The tubes undergo trimming, threading, and enamel coating, followed by drying and printing. Capping operations complete the process.

Advantages: i) Tamper-Proof: Cannot be refilled or reused. ii) Protective: Ensures contents remain free from contamination. iii) Elegant: Attractive appearance with a bright surface. iv) Non-toxic and Hygienic: Imparts no taste, flavor, odor, or color to contents. v) Light Weight: Reduces transport and handling costs. vi) Complete Collapsibility: Highly ductile for complete collapsibility. vii) Easy Availability: Increased aluminum production ensures raw material availability. viii) Economical: Unbreakable, easy to use, and carry.

Applications: a) Industrial Products: Adhesives, artist's colors, paints, etc. b) Cosmetics and Toiletries: Creams, shampoos, deodorants, toothpaste, etc. c) Food Products: Chutney, condensed milk, honey, etc. d) Household Items: Detergents, insecticides, shoe polish, etc. e) Pharmaceuticals: Antiseptic cream, ointments, veterinary creams, etc.

II. Multilayer Collapsible Plastic Tube (Lamitube)

Manufacturing Process: Multilayer collapsible plastic tubes, known as 'Lamitubes,' are produced by extruding a continuous tube and cutting it into the required length. The tube shoulder is compression molded, welded in a line operation, and printed separately in multicolor graphics. A lacquer coat protects the surface and provides the necessary finish.

Applications:

- Used for cosmetic products like lotions, creams, sprays, or liquids.
- In toiletry products like toothpaste, shaving cream, etc.
- For industrial products like greases, varnishes, and automotive creams.

<u>Composite Containers</u>

- A composite container, also known as Combican, is a versatile canister made from multiple materials, typically including paper, boards, and kraft papers, often with metal or plastic ends. In the case of a laminated composite container, kraft papers and boards are laminated with polyethylene or aluminum, creating a heat-sealed membrane that ensures complete leakproofing. This unique heat-sealing property enhances aroma retention and reduces moisture vapor permeability.

- **Applications:**

    - Food Products: Custard powder, masala powder, etc.
    - Detergents and Detergent Powders
    - Pharmaceutical Products: Tablets.
    - Pesticides.

- **New Developments:** Recent developments in leak-proof composite containers involve designing inner layers that enable double seaming of the cans, making them suitable for

packing liquid products. These innovations extend the use of composite containers to various items such as fruit juice, fruit pulp, preservatives, lube oil, motor oil, tooth powder, talcum powder, etc.

- **Advantages:** i) Alternative to Tin Plate Containers: Acts as an alternative packaging material to tin plate containers. ii) Disposable and Shelf-Stable: Suitable for goods with a shelf life of 1-2 years and offers disposable packaging. iii) Environment-Friendly: Composite containers are environmentally friendly. iv) Stability: Comparable stability to tin or glass containers for volumes up to one liter. v) Efficient Seaming: Seaming or sealing can be done on the same line as tin plate containers, ensuring efficiency in the packaging process.

Types of Extrusion process: Dry and Wet, Melting process

Dry Extrusion:

Process:

- In dry extrusion, the raw materials are processed in a solid state without the addition of external liquids. The materials are typically in the form of powders or granules.

Applications:

- **Food Industry:** Dry extrusion is widely used in the food industry for the production of various snacks, breakfast cereals, and pet food. It allows for the shaping and cooking of dry ingredients to create a range of textured and flavored products.
- **Plastics Industry:** In the plastics industry, dry extrusion is employed for processing thermoplastic materials. The solid plastic pellets or powder are melted and shaped into various forms, such as sheets, pipes, and profiles.

Advantages:

1. **Energy Efficiency:** Dry extrusion often requires less energy compared to wet processes since there is no need to evaporate water.
2. **Product Versatility:** It can be used for a wide range of materials, including those that are heat-sensitive or moisture-sensitive.
3. **Reduced Drying Time:** The absence of added moisture eliminates the need for drying, reducing processing time.

Disadvantages:

1. **Limited for Moisture-Sensitive Materials:** Some materials, especially those sensitive to heat or moisture, may not be suitable for dry extrusion.
2. **Texture Challenges:** Achieving certain textures, such as those that require moisture content, can be challenging.
3. **Equipment Wear:** The absence of lubricating effects from moisture might lead to increased wear and tear on equipment components.

Examples:

- Dry extrusion is commonly used in the production of extruded snacks like puffed corn snacks, cereals, and certain types of dry pet foods.

Definition: Wet extrusion is a food processing technique where a mixture of wet ingredients, often in the form of a dough or slurry, is pushed through a small opening (die) under high pressure to produce a specific shape. The wet nature of the ingredients distinguishes it from traditional dry extrusion processes.

Process:

1. **Preparation of Wet Mixture:** Ingredients are mixed with water or other liquids to form a wet, semi-fluid mixture.
2. **Forcing Through a Die:** The wet mixture is then forced through a die using a screw or piston mechanism.
3. **Shape Formation:** As the mixture passes through the die, it takes on the shape of the die, creating a continuous, uniform product.
4. **Cutting/Forming:** The extrudate is cut or shaped into the desired size before further processing or drying.

Working:

- The wet extrusion process relies on the pressure generated by a rotating screw or a piston to push the wet mixture through the die.
- The die plays a crucial role in determining the final shape and size of the extruded product.

Applications:

- **Snack Foods:** Wet extrusion is commonly used to produce various snack foods, such as extruded snacks, breakfast cereals, and pet treats.
- **Pasta Production:** Some types of pasta are produced using wet extrusion.
- **Meat Alternatives:** Wet extrusion is employed in the production of plant-based meat alternatives.
- **Baby Foods:** Formulations for baby foods often involve wet extrusion for shaping and processing.

Merits:

- **Versatility:** Can handle a wide range of ingredients and formulations.
- **Continuous Process:** Suitable for mass production with a continuous flow of output.
- **Shape Control:** Enables the creation of complex shapes and textures.

Demerits:

- **Equipment Complexity:** The machinery involved can be complex and requires careful maintenance.
- **Energy Consumption:** The process may require significant energy input.
- **Ingredient Sensitivity:** Some formulations may be sensitive to the extrusion process, affecting product quality.

Examples:

1. **Extruded Snacks:** Cheetos and other puffed snacks are often produced through wet extrusion.
2. **Breakfast Cereals:** Many types of breakfast cereals, such as certain types of granola, are manufactured using wet extrusion.
3. **Plant-Based Meat Alternatives:** Some plant-based meat products use wet extrusion for texture and shape.

Melt Extrusion:

Definition: Melt extrusion is a process in which a mixture of materials, often in the form of a polymer or a combination of polymers, is heated until it becomes molten and is then forced through a die to create a specific shape. While this term is more commonly associated with plastics and pharmaceuticals, it has some applications in the food industry as well.

Process:

1. **Melting:** The raw materials, which could include polymers, lipids, or other ingredients, are heated until they melt or become molten.
2. **Forcing Through a Die:** The molten material is then forced through a die to shape it into the desired form.
3. **Cooling/Solidification:** The extruded material is cooled to solidify and maintain its shape.

Working:

- Melt extrusion relies on the principles of heating a material until it reaches a molten state and then using pressure to force it through a die to create a specific shape.
- The temperature, pressure, and the choice of materials play a crucial role in determining the properties of the final extrudate.

Applications:

- **Pharmaceuticals:** Melt extrusion is extensively used in pharmaceuticals for drug delivery systems, where it helps to create controlled-release formulations.
- **Plastics Industry:** The most common application is in the production of plastic products and packaging.
- **Food Industry:** In the food industry, melt extrusion may be used for specific applications such as the production of certain types of confectionery or textured vegetable proteins.

Merits:

- **Uniformity:** Melt extrusion can produce materials with consistent properties.
- **Controlled Release:** In pharmaceutical applications, it allows for the controlled release of drugs.
- **Efficiency:** It is often a continuous and efficient process for mass production.

Demerits:

- **Equipment Complexity:** The machinery involved can be complex and requires careful maintenance.
- **Limited Heat-Sensitive Ingredients:** Not suitable for all food ingredients, especially those sensitive to high temperatures.
- **Energy Consumption:** The process may require significant energy input.

Examples:

1. **Pharmaceutical Tablets:** Melt extrusion is used to create drug formulations in the form of tablets for controlled drug release.
2. **Plastic Packaging:** Various types of plastic packaging materials are produced using melt extrusion processes.
3. **Certain Confectionery Products:** Some confectionery items may involve melt extrusion for shaping and forming.

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